ISE 350 Milestone 1

Hardware Integration & Active Safety

Inbound Signals (From Other Subsystems into Safety & Hardware)

Signal 1: new_speed

- Signal Name: new_speed
- Source Subsystem: Autonomous Vehicle Control
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Demanded forward velocity command generated by the speed controller.
- Data Type: single (or double)
- Units: m/s
- Expected Range: 0.0 to 3.5 m/s
- Update Frequency: 20 ms, 50 Hz
- Coordinate / Reference Frame: Vehicle Frame (ISO 8855, +X forward)
- Invalid / Fault Value: NaN, < 0.0 or > 3.5
- Behavior if Signal is Unavailable: Clamp commanded speed to 0 m/s within 100 ms (smooth down to stop).

Signal 2: steering_angle

- Signal Name: steering_angle
- Source Subsystem: Autonomous Vehicle Control
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Demanded road wheel steering angle generated by the path tracking controller.
- Data Type: single (or double)
- Units: rad/s or degrees?
- Expected Range: Approx 25 degrees (-0.436 to 0.436 rad)
- Update Frequency: 50 Hz,
- Coordinate / Reference Frame: Virtual single front wheel (CCW / Left turn = positive)
- Invalid / Fault Value: NaN, <-0.5 or >+0.5

- Behavior if Signal is Unavailable: Neutral lock, set to 0 rad, wheels commanded straight).

Signal 3: aeb_trigger_request

- Signal Name: aeb_trigger_request
- Source Subsystem: Perception & Sensor Fusion
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Discrete safety trigger flag asserted when perception detects an unavoidable forward collision.
- Data Type: boolean
- Units: N/A
- Expected Range: true or false
- Update Frequency: 20 Hz, 50 ms
- Coordinate / Reference Frame: Vehicle Frame
- Invalid / Fault Value: Non-boolean / undefined value
- Behavior if Signal is Unavailable: Safety supervisor defaults to its internal independent TTC estimation based on raw distance telemetry.

Signal 4: obstacle_ttc

- Signal Name: obstacle_ttc
- Source Subsystem: Perception & Sensor Fusion
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Continuous calculated Time-to-Collision to the primary forward tracked obstacle in the current travel corridor
- Data Type: single
- Units: seconds
- Expected Range: 0 to 10s
- Update Frequency: 20Hz, 50ms
- Coordinate / Reference Frame: Vehicle Frame (+X forward projection)
- Invalid / Fault Value: less than or equal to 0s while distance or NaN
- Behavior if Signal is Unavailable: Rely on auxiliary ultrasonic/bumper proximity sensor????; trigger AEB intervention if distance is less than critical safety distance

Signal 5: planner_heartbeat

- Signal Name: planner_heartbeat

- Source Subsystem: Motion Planning
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Incrementing alive counter monitored by the safety watchdog to confirm the trajectory planner has not stalled or crashed.
- Data Type: uint32
- Units: Counter ticks
- Expected Range: 0 to 2^32 – 1 (increments by +1 every execution cycle)
- Update Frequency: 20 Hz, 50 ms
- Coordinate / Reference Frame: N/A
- Invalid / Fault Value: Stalled counter value (change in count is zero for longer than 150 ms?)
- Behavior if Signal is Unavailable: Declare upstream planner failure, initiate controlled deceleration to full stop, and lock steering.

Outbound Signals (From Safety & Hardware to Other Subsystems)

Signal 6: fbk_wheel_speed

- Signal Name: fbk_wheel_speed (feedback wheel speed)
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Perception & Sensor Fusion
- Description: Measured vehicle ground speed derived from optical or Hall-effect wheel encoder pulse frequencies.
- Data Type: single
- Units: m/s
- Expected Range: 0.0 to 4.5 m/s
- Update Frequency: 100 Hz, 10ms
- Coordinate / Reference Frame: Longitudinal chassis axis (+X forward)
- Invalid / Fault Value: NaN, < 0.0, or pulse-rate accumulator overflow
- Behavior if Signal is Unavailable: Control and Perception fall back to visual/optical or motor open-loop back-EMF estimates?

Signal 7: fbk_actual_steering

- Signal Name: fbk_actual_steering
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control

- Description: Modeled actual road wheel steering angle accounting for servo actuation dynamics, calibration offsets, and mechanical rate limits.
- Data Type: single
- Units: rad or degrees
- Expected Range: Approx 25 degrees (-0.436 to 0.436 rad)
- Update Frequency: 50 Hz, 20 ms
- Coordinate / Reference Frame: Front axle center (CCW / Left turn = positive)
- Invalid / Fault Value: NaN,
- Behavior if Signal is Unavailable: Control assumes an idealized zero-lag actuator response model.

Signal 8: safety_supervisory_state

- Signal Name: safety_supervisory_state
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Motion Planning, Perception & Sensor Fusion
- Description: Current operating state of the Stateflow safety supervisor (0 = SYS_INIT_STANDBY, 1 = SYS_NOMINAL_ACTIVE, 2 = SYS_AEB_INTERVENTION, 3 = SYS_ESTOP_LATCHED).
- Data Type: uint8 (Enum)
- Units: N/A (State Enumeration)
- Expected Range: 0 to 3
- Update Frequency: 100 Hz, 10ms
- Coordinate / Reference Frame: N/A
- Invalid / Fault Value: 255 (STATE_FAULT) or any unmapped integer
- Behavior if Signal is Unavailable: Upstream nodes treat safety supervisor as offline and immediately suspend autonomous driving modes.

Signal 9: hw_hardware_health

- Signal Name: hw_hardware_health
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Motion Planning, Perception & Sensor Fusion

- Description: 16-bit status word containing diagnostic health flags for low-level electronics (Bit 0: E-Stop Tripped, Bit 1: Low Battery, Bit 2: ESC Fault, Bit 3: Steering Stall, Bit 4: Watchdog Timeout).
- Data Type: uint16 (Bitmask)
- Units: N/A (Binary bit flags)
- Expected Range: 0x0000 to 0xFFFF
- Update Frequency: 10 Hz, 100 ms
- Coordinate / Reference Frame: N/A
- Invalid / Fault Value: Non-zero critical fault bits asserted
- Behavior if Signal is Unavailable: Upstream software halts trajectory execution and signals operator intervention.

Signal 10: battery_rail_voltage

- Signal Name: battery_rail_voltage
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Master Telemetry / All Teams
- Description: Filtered DC terminal voltage of the main 2S LiPo drive battery measured via an ADC resistive divider
- Data Type: single
- Units: V
- Expected Range: 6.0 to 8.4
- Update Frequency: 10 Hz, 100 ms
- Coordinate / Reference Frame: Battery terminal reference to common chassis ground
- Invalid / Fault Value: < 6.4 V (critical under-voltage limit) or >8.6 V (over-voltage / sensor disconnect)
- Behavior if Signal is Unavailable: Issue soft operator warning at 6.8 V, automatically command parking brake stop at 6.4 V to prevent cell degradation and brownout crashes.

Signal 11: emergency_stop

- Signal Name: emergency_stop
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Master Telemetry / All Teams
- Description:

- Data Type:
- Units:
- Expected Range:
- Update Frequency:
- Coordinate / Reference Frame:
- Invalid / Fault Value:
- Behavior if Signal is Unavailable:

Team Responsibilities

Team Member	Initial responsibilities
Colin Iorio	Physical chassis assembly, hardware selection, and testing
Jessica Horncastle	Interface/data management, subsystem communication, and team coordination
Stephen Marth	Automatic Emergency Braking (AEB) programming and safety logic

- Communicate with all three other subsystem teams because our subsystem connects the software commands to the physical vehicle and handles safety overrides.

Team Dependencies

Perception & Sensor Fusion

What exact obstacle-distance and TTC signals will you send us?	
What units and data types will those signals use?	
How often will the obstacle information be updated?	
How will you indicate that sensor information is invalid or unavailable?	
Will Perception determine when AEB should trigger, or should our Safety subsystem make that decision from distance/TTC?	
What TTC or obstacle-distance values should be considered dangerous?	

Autonomous Vehicle Control

What exact speed command will be sent to us?	
What exact steering command will be sent to us?	
What units and ranges will be used for speed and steering?	
What update rate should we expect for these commands?	
What should the hardware do if a command becomes invalid or stops updating?	
When AEB or an emergency stop occurs, do you want a safety/status signal returned to Control?	
How should Control resume normal operation after the safety system has stopped the vehicle?	

Motion Planning

What safety information does Motion Planning need from us?	
Should we send a simple emergency-stop flag or a full safety state such as Normal, AEB, Emergency Stop, or Fault?	
How often does Motion Planning need the safety-state update?	
What should Motion Planning do when the vehicle enters AEB or safety lockout?	
Does Motion Planning need to send us a heartbeat or other signal so we can detect if the planner stops responding?	
What conditions should allow Motion Planning to begin sending movement commands again after a safety event?	

Test 1 – Date: TBD

Description: The first test will be performed in MATLAB/Simulink using simulated vehicle and obstacle data. The goal is to verify that the Active Safety subsystem correctly transitions from normal operation to an AEB or emergency-stop state when Time to Collision or obstacle distance reaches an unsafe threshold. We will verify that normal commands are allowed during safe conditions and that unsafe conditions override the normal vehicle command and produce a stop command.

Pass Criteria

- ✓ System stays in Normal mode during safe conditions
- ✓ Normal vehicle command passes through when TTC is safe
- ✓ AEB activates when TTC crosses the unsafe threshold
- ✓ Safety state changes correctly
- ✓ Normal → AEB/Emergency
- ✓ Normal speed command is overridden
- ✓ Stop command is produced
- ✓ Example: output speed goes to 0 m/s
- ✓ AEB activates without unexpected oscillation or state switching
- ✓ Invalid sensor input does not allow unsafe normal operation
- ✓ The correct response is visible in Simulink scopes/logs
- ✓ The response happens within the team's chosen maximum delay
- ✓ This can later become your measurable AEB response latency
- ✓ Repeated runs give the same expected result

Before we Test

- ✓ Basic Simulink/Stateflow safety model
  - Normal state
  - AEB/Emergency state
- ✓ Simulated vehicle speed input
- ✓ Simulated obstacle distance and/or TTC input
- ✓ Sensor-valid input
  - Valid / invalid
- ✓ A chosen AEB trigger threshold

- Example: TTC < ___ seconds

✓ A defined safe condition

- Example: TTC >___ seconds

✓ A defined unsafe condition

- Example: TTC between trigger and safe

✓ A stop output

- Example: commanded speed = 0 m/s

✓ A safety-state output

- Example: Normal / AEB / Emergency

✓ Scope or logging blocks

- So we can see TTC, speed command, and safety state during the test

✓ Agreement with Perception on what signal is being simulated

- TTC, obstacle distance, or both

✓ Agreement with Vehicle Control on what stop command should look like

Stephen notes

- What TTC threshold triggers AEB?
  Should It depends on speed? But for testing, we can set any condition to confirm logic.
- Should distance be a backup trigger?
  Should we also input distance with TTC to verify source and confirm emergency action? for testing we do not need to. But it could be another safety added.
- Who calculates TTC Perception or Safety?
- What is the maximum acceptable age of sensor data?
  Do we need one every sec or else emergency braking?
- Does AEB command `0 m/s`, direct braking/PWM, or both?
  Depends, for now I'm thinking binary output 0,1, this activates the AEB system. The system in the physical sense could be turning of the motors or an actual brake. But for now a signal that shows the state of EAB is good.
- What happens to steering during emergency braking?
  Should it straighten out or freeze in a position?
- Does AEB latch until manual reset?
  Do we revert states when we are within a safe speed at a safe distance (TTC), or do we reset by hitting a switch?
- What response-time requirement are we targeting?
  I'm not sure if this is so important until we have physical component implementation. The goal is understanding systems; the car could go 1 mph.
- What conditions define a successful AEB test?
  Taking in input, apply to condition, calculate, determine state, if state good, continue to monitor AEB off, if state bad, activate AEB while continuing to monitor, calculate state, if good turn off AEB while monitor state.

Questions?

When do y'all want to test for test1?